Prompted Segmentation for Drywall QA

1. Goal

The idea here is simple: given a drywall image and a short text prompt like "segment crack" or "segment taping area", the model should output a binary mask highlighting where the defect is. This covers two defect types — wall cracks and drywall joint taping — so the system could eventually plug into an automated inspection pipeline on a construction site.

2. Methodology

2.1 Why CLIPSeg

I went with CLIPSeg (`CIDAS/clipseg-rd64-refined`). It's built on top of CLIP's ViT-B/16 backbone and adds a lightweight decoder that uses FiLM conditioning to go from text + image → mask. One forward pass, done.

I looked at a few other options before settling on this:

Model	Text-Conditioned	Fine-tunable	Speed	Why Not
SAM + CLIP routing	Indirect	Two-stage hack	Moderate	No joint optimization — you'd be chaining two models and hoping the routing works
GroundingDINO + SAM	Yes	Limited	Slow	Way overkill for binary segmentation
LLaVA / BLIP-2 + decoder	Yes	No	Very slow	Needs way more GPU than what's available here

CLIPSeg won because it does text-to-mask natively, I can fine-tune just the decoder without blowing up VRAM, and the whole thing is a single forward pass which makes evaluation straightforward.

2.2 Training Setup

I only fine-tuned the decoder — 1.1M params, which is 0.7% of the whole model. The CLIP backbone stayed frozen. No point messing with pretrained vision-language features when VRAM is tight.

Component	Status	Params
CLIP backbone (vision + text)	Frozen	149.6M
Decoder (FiLM, projection, transformer, conv)	Trainable	1.1M

Prompt variants: I sampled random prompt phrasings per batch during training. This forces the model to not overfit to one exact wording: - Cracks: `"segment crack"`, `"segment wall crack"` - Taping: `"segment taping area"`, `"segment joint/tape"`, `"segment drywall seam"`

Loss: BCE + Dice, equal weight. BCE gives stable pixel-level gradients; Dice directly optimizes overlap, which is what we're actually measuring at eval time.

Optimizer: AdamW, lr=5e-5, weight decay 1e-4. Cosine LR schedule with 100-step linear warmup.

Augmentation (train only): Kept it conservative. Thin cracks fall apart under aggressive augmentation — seen it happen before. - HorizontalFlip (p=0.5) - RandomBrightnessContrast (±0.2, p=0.5) - GaussNoise (std 10–50, p=0.3) - RandomScale (±20%, p=0.3)

Inference: CLIPSeg's raw logits aren't calibrated on construction imagery, so I swept thresholds [0.2–0.7] on the validation set per prompt and picked the one with best mIoU. Logits get upsampled from 352×352 to 640×640 before thresholding — doing it the other way around introduces aliasing. Tried morphological closing too (3×3 kernel); kept it since it marginally helped.

3. Data Preparation

3.1 The Datasets

Both from Roboflow Universe, both already at 640×640, both in COCO JSON format:

Dataset	Images	Annotation Type	Source
Cracks	5,369	Polygon segmentation	Roboflow
Taping	1,022	Bounding box only	Roboflow

3.2 Splits

Roboflow exports already come pre-split (~70/15/15). I used them as-is — re-splitting would've been extra work for no real benefit.

Split	Cracks	Taping	Total
Train	3,758	715	4,473

Split	Cracks	Taping	Total
Valid	805	153	958
Test	806	154	960
Total	**5,369**	**1,022**	**6,391**

3.3 Mask Generation

COCO JSON → binary PNG masks ({0, 255}, single-channel): - **Cracks:** `cv2.fillPoly` on the polygon segmentation. Had to skip 16 annotations that had empty segmentation arrays (42 total across splits — not a big deal). - **Taping:** This is the tricky one. Every single annotation is a bounding box — no polygons at all. So I rasterized them as filled rectangles with `cv2.rectangle`. It's not perfect (real tape edges aren't perfectly rectangular), but it's a reasonable geometric proxy and it's all we have.

Seed fixed to `42` everywhere — augmentation, prompt sampling, everything.

4. Results

4.1 Test Metrics

Prompt	Threshold	Test mIoU	Test Dice
segment crack	0.4	0.4859	0.6338
segment taping area	0.4	0.5586	0.7063
Mean	**0.4**	**0.5222**	**0.6700**

4.2 Threshold Sweep (Validation Set)

Threshold	Cracks mIoU	Cracks Dice	Taping mIoU	Taping Dice
0.2	0.4937	0.6396	0.5605	0.7039
0.3	0.4975	0.6440	0.5701	0.7124
0.4	**0.4979**	**0.6451**	**0.5711**	**0.7136**
0.5	0.4962	0.6443	0.5685	0.7118
0.6	0.4922	0.6412	0.5629	0.7076
0.7	0.4851	0.6352	0.5518	0.6989

Both prompts peak at 0.4. The cracks curve is pretty flat (range 0.485–0.498), so threshold doesn't matter much there. Taping is more sensitive (0.552–0.571).

4.3 Consistency

Split each dataset into Low/Medium/High tertiles based on image brightness, edge density, and defect size. Here's how mIoU holds up:

Cracks:

Stratum	Low	Medium	High	Stable?
Brightness	0.4770	0.5018	0.4788	Yes
Edge density	0.4974	0.4888	0.4714	Yes
Defect size	0.3235	0.5429	0.5916	**No**

Taping:

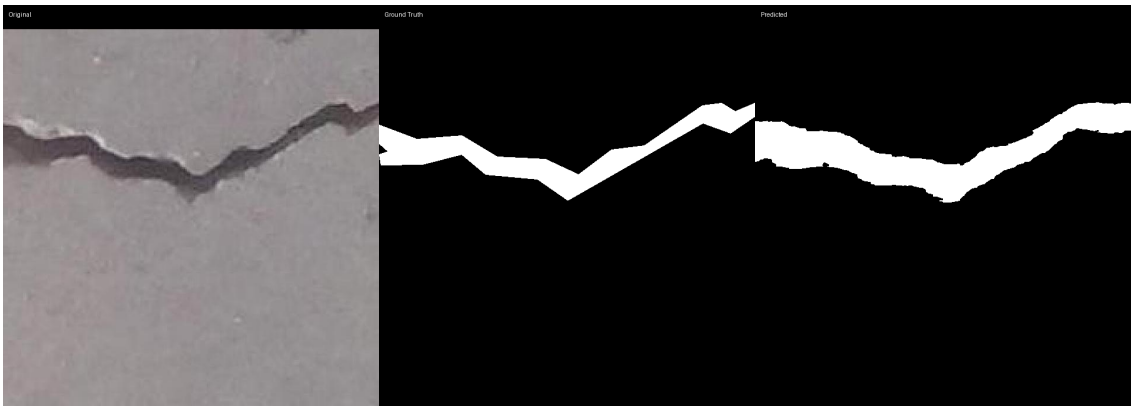
Stratum	Low	Medium	High	Stable?
Brightness	0.5634	0.5857	0.5265	Yes
Edge density	0.5895	0.5121	0.5736	Yes
Defect size	0.4966	0.5736	0.6068	Mostly

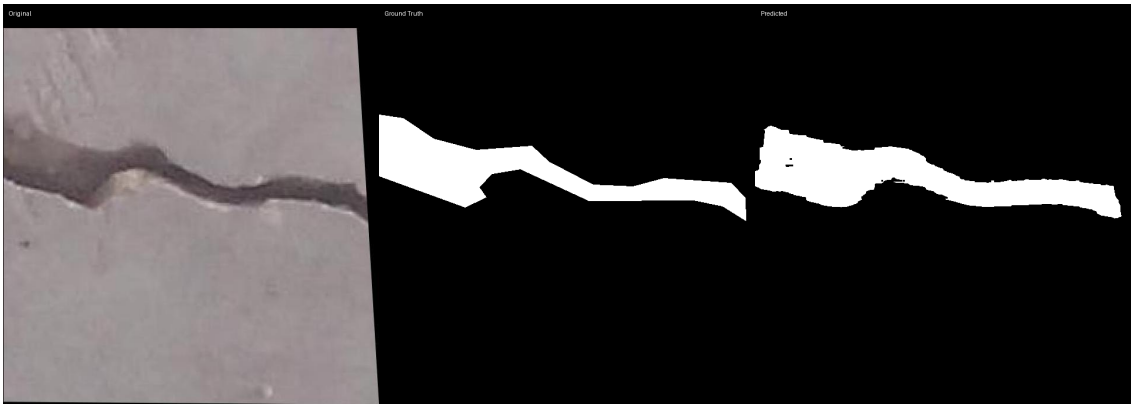
The model handles lighting and scene complexity fine. The real problem is defect size — small cracks (0.32 mIoU) get crushed compared to large ones (0.59). More on this in the failure section.

4.4 Visual Examples

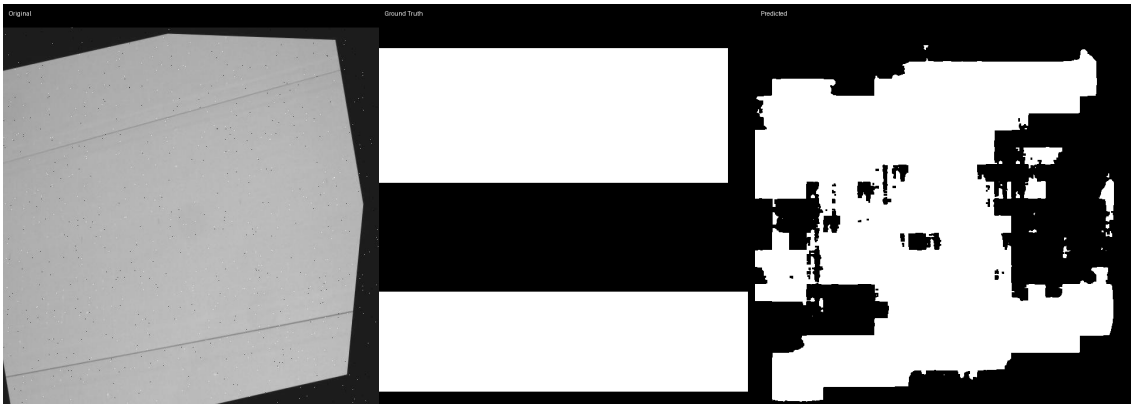
Original | Ground Truth | Predicted:

Cracks:





Taping:



4.5 Training Curve

20 epochs total, effective batch 4, cosine LR with 100-step warmup. Best checkpoint from epoch 17.

Epoch	Train Loss	Val mIoU (Cracks)	Val mIoU (Taping)	Mean mIoU
1	0.3664	0.4455	0.3582	0.4018
5	0.2825	0.4782	0.5205	0.4994
10	0.2631	0.4884	0.5580	0.5232

Epoch	Train Loss	Val mIoU (Cracks)	Val mIoU (Taping)	Mean mIoU
15	0.2541	0.4966	0.5660	0.5313
17	0.2538	0.4961	0.5678	**0.5319**
20	0.2531	0.4954	0.5668	0.5311

Model was basically done learning by epoch 15. The last 5 epochs barely moved the needle (0.5313→0.5319). The decoder-only approach hit its ceiling.

5. Failure Cases and What Could Fix Them

5.1 Small Cracks Get Missed

This is the biggest issue. Small/thin cracks score 0.32 mIoU while large ones hit 0.59. These are sub-5px-wide cracks that essentially disappear at CLIPSeg's 352×352 internal resolution — by the time you upsample back to 640, the detail is gone.

What might help: - Multi-scale inference — predict at 352 and 512, merge the outputs - CRF post-processing to sharpen thin boundaries - Bigger backbone (ViT-L), though that needs more VRAM

5.2 Taping GT Masks Are Just Rectangles

The taping dataset has zero polygon annotations. Every mask is a `cv2.rectangle` from the bounding box. Real tape seams aren't perfectly rectangular — they have angled or curved edges — so the GT itself is noisy. The model is being evaluated against an approximation, and there's a hard ceiling on how well it can do.

What might help: - Run SAM with bbox prompts to generate pseudo-polygon masks, then retrain on those - Manually fix a validation subset to measure how much the bbox proxy is hurting us

5.3 Model Capacity Runs Out

1.1M trainable params is not a lot. The model converged at epoch 15 and couldn't go further. The decoder architecture itself is the bottleneck — more training time won't help.

What might help: - Unfreeze the last few CLIP vision encoder blocks (~10M more params). Current VRAM usage is only 0.74GB out of 15.6GB, so there's plenty of headroom. - Knowledge distillation from a bigger model

6. Runtime and Footprint

Metric	Value
Total training time	~77 min (20 epochs on T4)
Avg inference time / image	~37 ms
Model checkpoint size	583.8 MB
Peak VRAM (training)	0.74 GB
Peak VRAM (inference)	0.74 GB
GPU	NVIDIA Tesla T4 (15.6 GB)

7. Reproducibility

- Seed fixed to `42` everywhere (PyTorch, NumPy, CUDA)
- Python 3.11, PyTorch 2.x (CUDA 12.x), transformers 4.x, albumentations 2.x
- `pip install -r requirements.txt`
- Train: `python train.py --batch_size 2 --grad_accum 2 --epochs 20`
- Infer: `python infer.py --checkpoint outputs/checkpoints/best.pt`
- Evaluate: `python evaluate.py --checkpoint outputs/checkpoints/best.pt`

8. Key Engineering Decisions

Decision	Choice	Why
Data split	Roboflow as-is	Already pre-split ~70/15/15; re-splitting adds complexity for no gain
Taping GT	Bbox → rectangle	No polygons exist; rectangle is a decent proxy for linear seams
Backbone	Frozen	VRAM budget; pretrained features are good enough
Decoder	Fully unfrozen	FiLM layers are key for text-to-mask alignment
Threshold	Per-prompt, swept on val	CLIPSeg logits aren't calibrated on this domain
Loss	BCE + Dice	BCE for stability, Dice matches the eval metric
Upsample	Before thresholding	Thresholding after upsample causes aliasing